Title: Web Server Deployment and Access Management using AWS EC2

- **Department:** Information Technology
- **Course Code:** 6IT1284
- **Course Name:** Hybrid Cloud Architecture and Management
- **Course Instructor:** Dr Durgadevi P
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Introduction

•Cloud computing allows users to deploy applications and services over the internet instead of local servers.
•Amazon Web Services (AWS) provides **Elastic Compute Cloud (EC2)** to create virtual machines in the cloud.
•A web server can be deployed on EC2 to host websites and web applications.
•Access management ensures that only authorized users can access and manage the server securely.
•This project demonstrates how to deploy a web server on AWS EC2 and configure secure access using cloud security features.

Basic Concept

What it is

Deployment of a web server on a cloud-based virtual machine using **AWS EC2** to host a website.

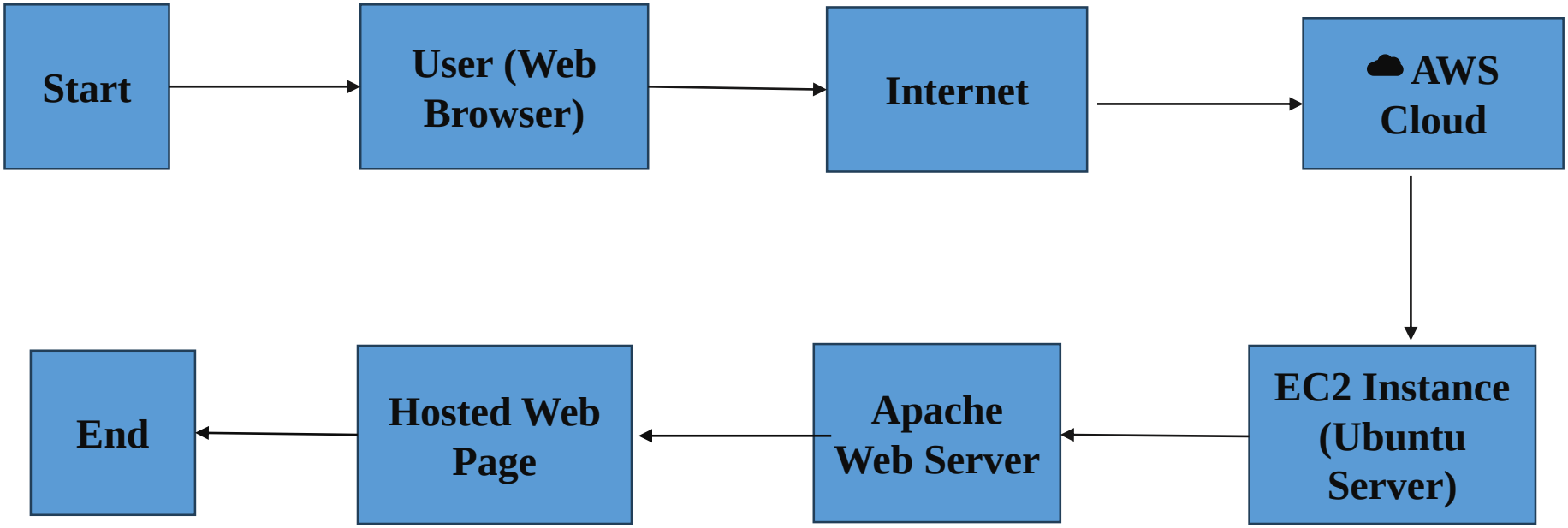
How it works

- Create an **EC2 instance** in AWS
- Install a **web server (Apache/Nginx)**
- Upload a web page
- Access it using the **public IP address**

Key Components

- AWS EC2 (Virtual Server)
- Web Server (Apache/Nginx)
- Security Groups
- SSH Key Pair

Architecture Diagram



Working Process

- User enters the **EC2 public IP address** in a web browser.
- The request travels through the **internet** to the **AWS cloud**.
- The request reaches the **EC2 virtual machine**.
- **Apache web server** processes the request.
- The **HTML webpage stored on the server** is retrieved.
- The webpage is **sent back to the user's browser** and displayed.

Real-World Example

- **Website Hosting** – Companies host their websites on cloud servers like AWS EC2.
- **E-commerce Platforms** – Online stores such as Amazon use cloud servers to handle user traffic.
- **Web Applications** – Platforms like Netflix and Spotify run their services on cloud infrastructure.
- **Company Websites** – Businesses deploy web servers in the cloud for global access.
- **Online Learning Platforms** – Educational websites use cloud servers to host content and applications.

Advantages

•**Scalability** – Resources can be increased or decreased based on demand.
•**Global Accessibility** – Websites can be accessed from anywhere through the internet.
•**Cost Effective** – Pay only for the resources used in the cloud.
•**Easy Deployment** – Web servers can be quickly created and managed.
•**High Availability** – Cloud providers ensure reliable and continuous service.
•**Secure Access Management** – Security groups control access to the server.

Conclusion

- Successfully deployed a **web server on AWS EC2**.
- Configured **security groups for secure access**.
- Installed **Apache web server** on the EC2 instance.
- Hosted a **webpage on the cloud server**.
- The website is **accessible through the public IP address**.
- This shows how **cloud computing can be used for web hosting and server management**.

References

Books

- Rajkumar Buyya – *Cloud Computing: Principles and Paradigms*
- Thomas Erl – *Cloud Computing: Concepts, Technology & Architecture*

Websites

- Amazon Web Services Official Documentation https://aws.amazon.com
- Apache HTTP Server Documentation – https://httpd.apache.org
- Ubuntu Server Guide –https://ubuntu.com/server/docs

Thank You